

EBN111 Lecture Schedule 2021

XY – Dr Xianming Ye; DLR – Dr Derik le Roux; SS – Dr Suzanne Smith; AL – Assistant Lecturers

Week	Day		Event	Tasks
1	Mo	15 March	Lecture 1	Watch video: Introduction and Study Guide Study Ch 1.1-1.4 and do textbook Examples: 1.1, 1.2, 1.3 Watch video: Units, Charge, Current, Voltage Do textbook Practice Problems: 1.1, 1.2, 1.3
	Tu	16 March	Lecture 2	Study Ch 1.5-1.6 and do textbook Examples: 1.4, 1.5, 1.6, 1.7 Watch video: Power, Energy and Circuit Elements Do textbook Practice Problems: 1.4, 1.5, 1.6, 1.7
	We	17 March	Lecture 3	Study Ch 2.2-2.3 and do textbook Examples: 2.1, 2.2, 2.3, 2.4 Watch video: Ohm's Law, Nodes & Branches & Loops Do textbook Practice Problems: 2.1, 2.2, 2.3, 2.4
	Th	18 March	Tut class # 1	Do chapter problems: 1.4, 1.6, 1.8, 1.11, 1.12, 1.16, 1.18, 1.20, 1.24, 1.28, 2.6, 2.7 Submit questions on ClickUP Discussions Forums.
	Fr	19 March		Watch and do AMS tutorial. Do AMS Assignment # 1 (Deadline: 23 March 2020 at 11:59pm).
2	Mo	22 March	Human Rights Day	
	Tu	23 March	Lecture 4	Study Ch 2.4 and do textbook Examples: 2.5, 2.6, 2.7, 2.8 Watch video: Kirchhoff's Law Do textbook Practice Problems: 2.5, 2.6, 2.7, 2.8
	We	24 March	Lecture 5	Study Ch 2.5-2.6 and do textbook Examples: 2.9, 2.10, 2.11, 2.12, 2.13 Watch video: Series and Voltage division, Parallel and Current division Do textbook Practice Problems: 2.9, 2.10, 2.11, 2.12, 2.13
	Th	25 March	Lecture 6 Monday Timetable	Study Ch 2.7 and do textbook Examples: 2.14, 2.15 Watch video and refer to slides for: Wye-Delta transformations Do textbook Practice Problems: 2.14, 2.15
	Fr	26 March	Tut class # 2	Do chapter problems: 2.15, 2.22, 2.23, 2.25, 2.32, 2.45b, 2.51b, 2.53b Submit questions on ClickUP Discussions Forums. Do AMS Assignment # 2 (Deadline: 30 March 2020 at 11:59pm)

Week	Day		Event	Tasks
3	Mo	29 March	Lecture 7	Study Ch 3.2 and do textbook Examples: 3.1, 3.2 Watch video: Nodal Analysis Do textbook Practice Problems: 3.1, 3.2
	Tu	30 March	Lecture 8	Study Ch 3.3 and do textbook Examples: 3.3, 3.4 Watch video: Nodal Analysis with Voltage Sources Do textbook Practice Problems: 3.3, 3.4
	We	31 March	Lecture on practicals	Pre-Practical 1
	Th	1 April	Tut class # 3	Do chapter problems: 3.2, 3.4, 3.12, 3.16, 3.20, 3.30, 3.31 Submit questions on ClickUP Discussions Forums
	Fr	2 April	Good Friday	
4	Mo	5 April	Family Day	
	Tu	6 April	Lecture 9	Study Ch 3.4 and do textbook Examples: 3.5, 3.6 Watch video: Mesh Analysis Do textbook Practice Problems: 3.5, 3.6
	We	7 April	Lecture 10	Study Ch 3.5 and do textbook Examples: 3.7 Watch video and refer to slides for: Mesh Analysis with Current Sources Do textbook Practice Problems: 3.7
	Th	8 April	Tut class # 4	Do chapter problems: 3.37, 3.44, 3.60 Submit questions on ClickUP Discussions Forums
	Fr	9 April		Do ClickUP Assignment # 3. (Deadline: 13 April 2020 at 11:59pm)
5 Practical 1	Mo	12 April	Class Test 1	Chapters 1, 2 & 3.2 – 3.5
	Tu	13 April	Lecture 11	Study Ch 3.6-3.7 and do textbook Examples: 3.8, 3.9 Watch video: Nodal and Mesh Analysis by Inspection Do textbook Practice Problems: 3.8, 3.9
	We	14 April	Lecture 12	Study Ch 3.9 and do textbook Examples: 3.12, 3.13 Watch video: Transistors Do textbook Practice Problems: 3.12, 3.13
	Th	15 April	Tut class # 5	Do chapter problems: 3.69, 3.72, 3.90, 3.91 Submit questions on ClickUP Discussions Forums.
	Fr	16 April		Do AMS Assignment # 4 (Deadline: 20 April 2020 at 11:59pm).

Week	Day		Event	Tasks
6	Mo – Fr 19 – 23-April		Test Week 1 Test covers Chapters 1-3; Marks: 45 ClickUP Announcements will be posted with further details.	
7	Mo	26 April	UP Holiday	
	Tu	27 April	Freedom Day	
	We	28 April	Lecture 13 Monday Timetable	Study Ch 4.2-4.3 and do textbook Examples: 4.1, 4.2, 4.3, 4.4, 4.5 Watch video and refer to slides for: Linearity and Superposition Do textbook Practice Problems: 4.1, 4.2, 4.3, 4.4, 4.5
	Th	29 April	Lecture 14	Study Ch 4.4 and do textbook Examples: 4.6, 4.7 Watch video: Source Transformation Do textbook Practice Problems: 4.6, 4.7
	Fr	30 April		View Semester Test feedback video
8	Mo	3 May	Lecture 15	Study Ch 4.5 and do textbook Examples: 4.8, 4.9, 4.10 Watch video: Thevenin's Theorem Do textbook Practice Problems: 4.8, 4.9, 4.10
	Tu	4 May	Lecture 16	Study Ch 4.6 and do textbook Examples: 4.11, 4.12 Watch video: Norton's Theorem Do textbook Practice Problems: 4.11, 4.12
	We	5 May	Lecture on practicals	Pre-Practical 2
	Th	6 May	Tut class # 6	Do chapter problems: 4.4, 4.5, 4.11, 4.18, 4.24, 4.37, 4.38, 4.40, 4.48, 4.52 Submit questions on ClickUP Discussions Forums
	Fr	7 May		

Week	Day		Event	Tasks
9	Mo	17 May	Lecture 17	Study Ch 4.8 and do textbook Examples: 4.13 Watch video: Maximum Power Transfer Do textbook Practice Problems: 4.13
	Tu	18 May	Lecture 18	Study Ch 5.2-5.3 and do textbook Examples: 5.1, 5.2 Watch video: Op Amps, Ideal Op Amps Do textbook Practice Problems: 5.1, 5.2
	We	19 May	Lecture 19	Study Ch 5.4-5.5 and do textbook Examples: 5.3, 5.4, 5.5 Watch video: Inverting and Non-inverting Amplifiers Do textbook Practice Problems: 5.3, 5.4, 5.5
	Th	20-May	Tut class # 7	Do chapter problems: 4.71, 4.72, 5.6, 5.8, 5.9, 5.13, 5.20. Submit questions on ClickUP Discussions Forums
	Fr	21 May		Do AMS Assignment # 5 (Deadline: 25 May 2020 at 11:59pm).
10 May – 14 May: UP holiday				
10 Practical 2	Mo	24 May	Lecture 20	Study Ch 5.6-5.7 and do textbook Examples: 5.6, 5.7, 5.8 Watch video: Summing and Difference Amplifiers Do textbook Practice Problems: 5.6, 5.7, 5.8
	Tu	25 May	Lecture 21	Study Ch 5.8 and do textbook Examples: 5.9, 5.10 Watch video: Cascade Circuits Do textbook Practice Problems: 5.9, 5.10
	We	26 May	Lecture 22	Study Ch 6.2-6.3 and do textbook Examples: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7 Watch video: Capacitors, Series and Parallel Capacitors Do textbook Practice Problems: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7
	Th	27 May	Tut class # 8	Do chapter problems: 5.31, 5.32, 5.39, 5.45, 5.57, 5.70 Submit any questions on ClickUP Discussions Forums
	Fr	28 May		Do AMS Assignment # 6 (Deadline: 20 April 2020 at 11:59pm).

Week	Day		Event	Tasks
11	Mo	31 May	Class Test 2	Chapters 4 & 5
	Tu	1 June	Lecture 23	Study Ch 6.4-6.5 and do textbook Examples: 6.8, 6.9, 6.10, 6.11, 6.12 Watch video: <i>Inductors, Series and Parallel Inductors</i> Do textbook Practice Problems: 6.8, 6.9, 6.10, 6.11, 6.12
	We	2 June	Lecture 24	Study Appendix B and do textbook Examples: B.1, B.2, B.3, B.4, B.5 Watch video: <i>Complex numbers</i> Do textbook Practice Problems: B.1, B.2, B.3, B.4, B.5
	Th	3 June	Tut class # 9	Do chapter problems: 6.10, 6.12, 6.18, 6.40, 6.46, 6.54 Submit any questions on ClickUP Discussions Forums
	Fr	4 June		Do AMS Assignment # 7 (Deadline: 8 June 2020 at 11:59pm).
12	Mo – Fr 7 – 11 June		Test Week 2 Test covers Chapters 4, 5 and 6 Marks: 45 ClickUP Announcements will be posted with further details.	
13	Mo	14 June	Lecture 25	Study Ch 9.2-9.3 and do textbook Examples: 9.1, 9.2 Watch video: <i>Sinusoids and Phasors</i> Do textbook Practice Problems: 9.1, 9.2 Watch Video: Pre-practical 3
	Tu	15 June	Lecture 26	Study Ch 9.3-9.4 and do textbook Examples: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8 Watch video: <i>Phasors and Phasor Relationships with Circuit Elements</i> Do textbook Practice Problems: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8
	We	16 June	Lecture 27	Study Ch 9.5-9.7 and do textbook Examples: 9.9, 9.10, 9.11, 9.12 Watch video: <i>Impedance, Kirchhoff's Laws in Frequency Domain</i> Do textbook Practice Problems: 9.9, 9.10, 9.11, 9.12
	Th	17 June	Tut class # 10	Do chapter problems: 9.2, 9.6, 9.13, 9.19, 9.20, 9.42, 9.48, 9.54, 9.66, 9.70 Submit any questions on ClickUP Discussions Forums
	Fr	18 June		Do AMS Assignment # 8 (Deadline: 24 June 2020 at 11:59pm). View Semester Test feedback video.

Week	Day		Event	Tasks
14 Practical 3	Mo	21 June	Lecture 28	Study Ch 10.2 and do textbook Examples: 10.1, 10.2 Watch video: Nodal Analysis Do textbook Practice Problems: 10.1, 10.2
	Tu	22 June	Lecture 29	Study Ch 10.3-10.4 and do textbook Examples: 10.3, 10.4, 10.5, 10.6 Watch video: Mesh Analysis, Superposition Do textbook Practice Problems: 10.3, 10.4, 10.5, 10.6
	We	23 June	Lecture 30	Study Ch 10.5-10.6 and do textbook Examples: 10.7, 10.8, 10.9, 10.10 Watch video: Source Transformation, Thevenin & Norton Do textbook Practice Problems: 10.7, 10.8, 10.9, 10.10
	Th	24 June	Tut class # 11	Do chapter problems: 10.15, 10.19, 10.36, 10.46, 10.52, 10.56, 10.61, 10.66, 10.68 Submit any questions on ClickUP Discussions Forums
	Fr	25 June		Do AMS Assignment # 9 (Deadline: 29 June 2020 at 11:59pm).
15	Mo	28 June	Class Test 2	Chapters 6 & 9
	Tu	29 June		
	We	30 June	Semester Test 3	Optional Semester Test
	Th	1 July		
	Fr	2 July		
16	Mo	5 July	Lecture 28	Wednesday Timetable: No lectures
	Tu	2 July	Lecture 29	UP Holiday
	We	3-June	Lecture 30	Exams start